PROPITHUB PLATFORM

PROPITHUB PLATFORM WORKFLOWS AUTOMATION RESEARCH REPORT & BLUEPRINT

Comprehensive Codebase Audit, Communications Vendor Economics (Meta Cloud vs. Twilio vs. Infobip), Serverless Architecture & Execution Roadmap

ARCHITECTURE & SPECIFICATION SCOPE

Document Version: 2.4.0-PROD | Classification: Formal Architectural Blueprint

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Target Roles (Unified App): Property Owner, Tenant Resident, Field Maintenance, Super Admin

Strictly Excluded Role: External Low-Code Tools (Zero n8n/Zapier Dependency) (Web Management Portal Only)

Zero Redundancy

Single Codebase

One unified application dynamically adapting UI, navigation, and features based on verified user credentials.

Hard RBAC Wall

Zero-Financial Manager

Strict role boundaries ensuring delegated Property Managers have full facility controls with zero financial exposure.

26 Modules

Complete Lifecycle

End-to-end coverage: Lease dispatch, digital signing, rent invoicing, cash claims, move-out settlement, technician dispatch.

This document contains the complete, authoritative product specification, business logic inventory, and mobile UX blueprints.

1 EXECUTIVE SUMMARY & FOUNDATIONAL PLATFORM REALITIES

This research document establishes the formal architectural strategy for automating operational communications across Propithub. It synthesizes an exhaustive audit of the active Propithub_v2 codebase (Firebase Cloud Functions, React/TypeScript Web, and Expo Mobile) with regional telecommunications laws, vendor pricing models, and serverless reliability engineering.

Fundamental Platform Realities Verified in Codebase

- **Physical-Only Lease Execution:** Propithub strictly enforces physical wet-ink lease execution. Tenancy contracts are printed, physically signed, and counter-signed. The platform tracks renewal countdowns, reminders, and archives PDF contracts. Workflows must NEVER mark a lease as executed algorithmically without human manager verification.
- **Zero Dispute Engine:** Unlike generic platforms, Propithub features zero automated dispute arbitration. Grievances and tenant-owner claims are strictly routed to human property managers for administrative resolution.
- **Existing Communications Infrastructure:** functions/src/index.ts already houses production-grade Twilio SMS (sendTwilioSms) and SendGrid Email (sendSendGridEmail, sendInvoiceEmail), managed via Firestore configuration at /platformConfig/comms. Automated midnight sweeps already run on Cloud Scheduler.
- **The WhatsApp Opportunity:** Outbound mobile messaging is currently 100% SMS. WhatsApp Business messaging represents an immediate 50% to 70% cost reduction over regional Middle Eastern SMS, combined with over 90% read rates within 15 minutes.
- **Architectural Verdict on Orchestration:** Firebase Cloud Functions is 100% sufficient. Introducing an external orchestrator (n8n, Make, or Zapier) introduces unacceptable security risks, latency, split-brain logic, and server maintenance overhead.

STRATEGIC RECOMMENDATION

Execute a Fast-Track Pilot on Twilio for WhatsApp (Rent Reminders & Receipts) leveraging existing backend infrastructure, while retaining Firebase Cloud Functions as the single source of truth.

2 COMPREHENSIVE AUDIT OF THE 8 TARGET WORKFLOWS

Every candidate workflow was audited against existing functions/src/index.ts triggers, React web pages, and mobile screens to identify current state, manual bottlenecks, and recommended channel mix.

Workflow	Current State	Manual Bottleneck	Trigger Event	Channel Mix	Priority
1. Rent Reminders	Automated (Cron) SMS + Email	Manager manual calls when SMS unread	Cloud Scheduler @ 00:00 daily	WhatsApp (Primary) Email (Backup)	HIGH (Strongest)
2. Payment Receipts	Automated (Trigger) SMS + Email	Manual cash entry verification	Firestore Trigger: invoices/{id} -> paid	WhatsApp (Summary) Email (Tax PDF)	HIGH (Essential)
3. Maintenance Updates	Partial in onTicketWrite (Emergency/Resolved)	Manager/tech manual phone calls	Firestore Trigger: tickets/{id} status	WhatsApp (Both) In-App Push	HIGH (Field Staff)
4. Lease Renewals	Partial in Cron (30, 15, 7 days)	Drafting & printing physical leases	Cloud Scheduler: T-60, T-30 days	Email (Draft) WhatsApp (Nudge)	MEDIUM (Operational)
5. Onboarding Invites	Missing Comms (Auth user created only)	Manager manually texts passwords	Auth / Firestore User Provisioning	WhatsApp (Invite) Email (Welcome Pack)	HIGH (Frictionless)
6. Announcements	In-App Only (SMS/Email skipped)	Landlords print paper bulletins	Manager clicks Publish Announcement	In-App (Primary) WhatsApp (Emergency)	LOW / CAUTION (Spam Risk)

7. Platform SaaS Billing	Automated in onPlatformInvoiceWrite	Super-admin manual subscription chase	Stripe/Razorpay Webhook failure	Email (Primary) In-App Modal	LOW (for WA) (B2B Desktop)
8. Super-Admin Alerts	Missing in Comms (Firestore logs only)	Manual inspection of error logs	Quota exceed / Error Cloud Function alert	Slack/Discord Webhook Email Alert	LOW (for WA) (DevOps Scope)

3 REGULATORY COMPLIANCE & HUMAN-IN-THE-LOOP SAFEGUARD

Regional Data Privacy & Messaging Laws

- **European Union (GDPR & ePrivacy):** Operational transactional messages (receipts, urgent repairs, rent due) are lawful under Contractual Necessity (Art. 6(1)(b) GDPR). Non-essential broadcasts require explicit opt-in checkboxes. An automated STOP keyword or profile toggle must immediately write whatsappOptOut: true to Firestore.
- **United Arab Emirates (TDRA & Decree-Law No. 45/2021):** Requires officially registered corporate sender IDs. Tenancy notifications tied to registered Ejari contracts are compliant, but unsolicited marketing carries fines up to AED 1,000,000.
- **Saudi Arabia (PDPL & CITC / CST):** Strict personal data handling and consent verification. System must upgrade hardcoded +91 country prefix in functions/src/index.ts:117 to dynamic E.164 normalization supporting KSA (+966), UAE (+971), and Europe.

Human-in-the-Loop Safeguards (Non-Negotiable Invariants)

INVARIANT 1: STRICTLY PHYSICAL LEASE SIGNATURES

Propithub does not utilize digital e-signatures. Automated workflows send reminders and generate PDF lease packs, but MUST NEVER automatically transition a lease to "active" without human property manager confirmation of a signed physical wet-ink document.

INVARIANT 2: INSPECTION DEDUCTIONS & MOVE-OUT SETTLEMENTS

Damage deductions can never be calculated algorithmically. A human property manager or inspector must record physical inspection notes before any deposit settlement statement is triggered.

INVARIANT 3: ZERO ALGORITHMIC DISPUTE ARBITRATION

Resident grievances or disputes must route directly to a property manager; automated closure or dispute judgment by software is prohibited.

4 RESOLUTION OF THE 6 OPEN RESEARCH QUESTIONS

Q1: What is already automated in the live app today?

Cloud Scheduler runs dispatchDailyRentReminders and applyLateFeesDaily every 24 hours. Firestore database triggers handle onRentalInvoiceWrite (SMS/Email receipt), onAgreementWrite, and onTicketWrite (emergency/resolved only). Onboarding invites, mid-stage maintenance updates, and broadcast distributions remain 100% manual.

Q2: Which workflows genuinely need WhatsApp vs. Email vs. In-App?

WhatsApp is reserved for high-urgency operational alerts requiring immediate action (rent reminders with payment links, payment confirmation summaries, technician arrival alerts, and account activation). Email is reserved for legal, dense, or archival documents (official PDF tax invoices, tenancy renewal packets, formal deposit settlements). In-App handles non-urgent community rules and bulletins.

Q3: Does a payment receipt need to be a legal PDF or is chat confirmation enough?

A dual-tier approach is mandatory: Chat messages alone do not satisfy VAT/tax accounting standards in the UAE (FTA) or Saudi Arabia (ZATCA e-invoicing). The WhatsApp message delivers an immediate conversational summary with transaction metadata and an authenticated direct download link to the official, archived PDF tax invoice generated by Propithub.

Q4: Who authors newsletter content, and does platform or owner send it?

In OwnerMessagesPage.tsx, property owners author building-specific operational notices. If owners send mass broadcasts over a shared official Propithub WhatsApp number and tenants flag messages as spam, Meta downgrades or permanently bans the platform number, halting rent reminders and receipts for all landlords. Policy: Broadcasts remain In-App/Email; WhatsApp broadcasts are restricted to urgent emergencies with explicit attribution: "[Building Name] Notice: ...".

Q5: What are the projected messaging volumes and costs at scale?

At 1,000 tenants (~3,500 msgs/mo), Twilio SMS costs ~$280/mo vs. Twilio WhatsApp at ~$115/mo. At 10,000 tenants (~35,000 msgs/mo), Twilio SMS reaches ~$2,800/mo vs. Twilio WhatsApp at ~$1,150/mo. WhatsApp delivers a sustained 50% to 70% cost reduction over regional SMS in the GCC.

Q6: Which workflows require identity verification that chat cannot provide?

Bank and payout account modifications must NEVER be accepted over WhatsApp chat due to SIM-swap vulnerabilities. Tenancy contract acknowledgments must occur inside authenticated Propithub web or mobile sessions. WhatsApp serves purely as the secure deep-link transport layer.

5 TECHNOLOGY STACK, COST ECONOMICS & ARCHITECTURE EVAL

A. Direct Meta Cloud API vs. Twilio for WhatsApp

Every WhatsApp Business notification incurs a Meta base fee by country and category (Utility). Meta Direct passes wholesale rates with $0 markup. Twilio passes Meta base fees and adds a +$0.0050/msg surcharge.

Country / Calling Code	Meta Base (Direct)	Twilio Surcharge	Total via Twilio	Twilio Markup Impact
India (+91)	$0.0042 (~Rs 0.35)	+$0.0050	$0.0092 (~Rs 0.77)	+119% (More than double)
UAE (+971)	$0.0335	+$0.0050	$0.0385	+14.9% surcharge
Saudi Arabia (+966)	$0.0382	+$0.0050	$0.0432	+13.1% surcharge
United Kingdom (+44)	$0.0390	+$0.0050	$0.0440	+12.8% surcharge
United States (+1)	$0.0150	+$0.0050	$0.0200	+33.3% surcharge

Monthly Cost Scaling at Volume Breakpoints

Monthly Volume	Direct Meta Cloud API	Twilio for WhatsApp	Twilio Markup Surcharge	Strategic Recommendation
1,000 msgs	~$35 / mo	~$40 / mo	+$5 / mo	Twilio markup negligible; rapid launch.
5,000 msgs	~$175 / mo	~$200 / mo	+$25 / mo	Twilio strongly recommended; saves dev ops.
10,000 msgs	~$350 / mo	~$400 / mo	+$50 / mo	Twilio remains highly cost-effective.
25,000 msgs	~$875 / mo	~$1,000 / mo	+$125 / mo	Inflection point: plan Direct Meta migration.

50,000 msgs	~$1,750 / mo	~$2,000 / mo	+$250 / mo	Direct Meta saves $3,000/yr; migrate.
100,000 msgs	~$3,500 / mo	~$4,000 / mo	+$500 / mo	Direct Meta saves $6,000/yr; mandatory.

B. Evaluation of the 3 Industry Stacks

Metric / Component	Option A: Twilio All-in-One	Option B: Modern Best-of-Breed	Option C: Infobip Enterprise
Channel Mix	SendGrid + Twilio SMS + Twilio WA	Resend/Postmark + Twilio SMS + Meta	Infobip Email + SMS + WhatsApp
Base Fee	$0 (Pay-as-you-go)	$0 (Pay-as-you-go)	$150 - $300 / mo minimum commit
1k Tenants Cost	~$200 / mo (Email $20, SMS $80, WA $100)	~$188 / mo (Email $20, SMS $80, WA $88)	~$350 - $450 / mo (Driven by minimum)
5k Tenants Cost	~$840 / mo (Email $20, SMS $320, WA $500)	~$778 / mo (Email $20, SMS $320, WA $438)	~$850 - $950 / mo
10k Tenants Cost	~$1,660 / mo (Email $20, SMS $640, WA $1k)	~$1,535 / mo (Email $20, SMS $640, WA $875)	~$1,500 - $1,650 / mo
Engineering Setup	1 - 2 Days (Code already in repo)	1 - 2 Weeks (Meta custom app & hooks)	2 - 3 Weeks (Procurement & SDK setup)
Vendor Management	1 Vendor (Twilio single console)	3 Vendors (Twilio, Meta, Resend)	1 Enterprise Vendor (Custom contract)
Strategic Verdict	RECOMMENDED FOR PILOT & GROWTH	RECOMMENDED FOR LONG-TERM SCALE	RECOMMENDED ONLY FOR 50k+ GCC TENANTS

C. Architecture Evaluation: Is Firebase Enough or Do We Need n8n?

Definitive Verdict: Firebase Cloud Functions is 100% sufficient. Adding n8n or Zapier is strongly discouraged.

- **Hosting & Reliability:** Firebase is 100% serverless, scaling automatically with 99.95%+ Google Cloud SLA. n8n requires managing a dedicated Docker VPS ($10-$40/mo), creating a single point of failure if the container or Redis queue crashes.
- **Architectural Integrity:** Firebase maintains a single source of truth in Git (functions/src/). An external tool creates a fragile "split-brain" architecture where half the business rules live in code and half live on visual canvas nodes.
- **Security & Privacy:** Firebase communicates directly with Firestore using encrypted service credentials. n8n requires exporting private Firebase Admin service keys and exposing public unauthenticated webhook endpoints.
- **Latency & Cost:** Firebase executes sub-second (50-250ms) within the Google Cloud cluster. External webhooks introduce 1.5s to 4.0s network delays. Google Cloud includes 2,000,000 function invocations per month for free.

6 ARCHITECTURAL PROCESS PIPELINES

Pipeline 1: Rent Due Reminder & Instant Payment Receipt Flow

- **Step 1 (Daily Sweep):** Cloud Scheduler triggers dispatchDailyRentReminders daily at 00:00. Functions query unpaid invoices due in 5 days, today, or overdue.
- **Step 2 (WhatsApp Template Dispatch):** Dispatches Twilio WhatsApp Utility template (rent_due_reminder) with dynamic variables (Tenant Name, Property, Amount, Due Date, Pay Link).
- **Step 3 (Automatic SMS Fallback):** If Twilio returns a delivery failure (unregistered on WhatsApp or bounced), the function automatically catches the event and dispatches a plain Twilio SMS.
- **Step 4 (Online Settlement Trigger):** Tenant pays via payment gateway. Webhook updates invoice status to "paid". onRentalInvoiceWrite fires immediately.
- **Step 5 (Dual-Tier Receipt Dispatch):** Fires instant WhatsApp receipt summary with direct PDF link, while SendGrid dispatches the formal PDF tax invoice email.

Pipeline 2: Maintenance Ticket Lifecycle Updates

- **Step 1 (Ticket Created):** Tenant logs repair request in mobile app. onTicketWrite triggers WhatsApp alert to Property Manager: "New Ticket #104 in Unit 402".
- **Step 2 (Staff Assigned):** Manager assigns field technician. onTicketWrite triggers WhatsApp job dispatch to technician with unit details, and notifies tenant: "Technician Tariq assigned".
- **Step 3 (Work in Progress):** Technician marks ticket "in_progress". Tenant receives WhatsApp update: "Technician has started work".
- **Step 4 (Resolution & Proof):** Technician completes repair, uploads photo proof, and marks "resolved". Tenant receives WhatsApp completion notice with a direct 5-star service rating button. Manager receives summary notice.

7 IMPLEMENTATION SCENARIOS & ENGINEERING VELOCITY

Comparison of Implementation Scenarios

SCENARIO A: FAST-TRACK PILOT
3 - 5 Business Days (Twilio WhatsApp Reminders & Receip

SCENARIO B: FULL MULTI-WORKFLOW
7 - 10 Business Days (All 8 Workflows + Mobile/Web Preferences)

SCENARIO C: DIRECT META INTEGRATION
14 - 18 Business Days (Custom Meta Webhooks & SMS Fall

RECOMMENDED APPROACH
Deploy Scenario A immediately; scale into Scenario B over 1 week

Surface-by-Surface Engineering Scope

- **Backend Cloud Functions (functions/):** Add sendTwilioWhatsApp helper, dynamic E.164 phone parser, maintenance ticket state machine triggers, and onboarding invitation dispatcher. Estimated coding time: ~3 to 5 hours.
- **Web Management Portal (src/):** Superadmin WhatsApp feature toggles, tenant opt-in/opt-out status, urgent emergency broadcast guardrails, and public tax receipt landing page. Estimated coding time: ~3 to 4 hours.
- **Mobile Application (mobile/):** Notification channel preferences in AccountSettingsScreen.tsx and deep-link routing from WhatsApp into Invoice and Ticket screens. Estimated coding time: ~2 to 3 hours.

Development Velocity vs. Real-World Timeline

Phase / Component	AI Pair Dev Effort	External Dependency	Elapsed Duration
Backend & Cloud Functions	~4 hours	Zero (Direct local development)	Day 1
Web & Mobile Screens	~6 hours	User visual review & approval	Day 2
Twilio Sandbox Testing	~2 hours	User test messages on real mobile phone	Day 2 - Day 3
Meta Business Verification	N/A (Administrative)	Meta corporate document review	1 - 3 business days
Meta Template Approval	N/A (Console)	Meta automated AI review	15 mins - 2 hours
Full Production Launch	~1 hour	Final deployment (firebase deploy)	Day 5 - Day 7

8 TWILIO WHATSAPP SENDER & META APPROVAL GUIDE

Step 1: Required Verification Documents

- **Meta Business Account ID:** 15-digit ID from business.facebook.com (Settings -> Business Info).
- **Corporate Proof Document:** Trade License, Certificate of Incorporation, or Tax Certificate with exact matching legal address.
- **Dedicated Phone Number:** Cannot be currently active on the WhatsApp consumer or business mobile app; must receive verification SMS/call.
- **Live Corporate Website:** Must clearly display legal name, contact information, and privacy policy.

Step 2: Twilio Embedded Signup & Template Submission

Navigate to Twilio Console -> Messaging -> Senders -> WhatsApp senders -> New WhatsApp Sender. Authenticate with Facebook, select Meta Business Account, enter official display name ("Propithub"), and verify phone number via 6-digit SMS code.

Submit the 3 pre-approved Utility message templates under Content Template Builder:

- **Template 1: rent_due_reminder:** "Hello {{1}}, reminder from {{2}} that rent for Unit {{3}} (Amount: {{4}}) is due on {{5}}. Review invoice and pay securely: {{6}}"
- **Template 2: payment_received_receipt:** "Thank you {{1}}! Your rent payment of {{2}} for Unit {{3}} has been received. Receipt #{{4}}. Download official tax receipt: {{5}}"
- **Template 3: maintenance_ticket_update:** "Update on Ticket #{{1}} ({{2}}): Status is now {{3}}. Assigned technician: {{4}}. View progress: {{5}}"

> **PRO-TIP: IMMEDIATE SANDBOX TESTING**
> Do not wait for Meta verification to begin testing. Use Twilio WhatsApp Sandbox (Messaging -> Try it out) to connect your phone instantly on Day 1 and validate all automated functions.

9 RECOMMENDED ACTION PLAN & NEXT STEPS

- **Step 1 (Immediate Pilot):** Connect test device to Twilio Sandbox; implement sendTwilioWhatsApp in functions/src/index.ts with automatic SMS fallback.
- **Step 2 (Administrative Submission):** Submit Meta Business Manager verification documents and register the 3 Utility Content Templates in Twilio.
- **Step 3 (Client Settings):** Add notification preferences to mobile AccountSettingsScreen.tsx and emergency broadcast safeguards to OwnerMessagesPage.tsx.
- **Step 4 (Production Cutover):** Switch Twilio sender from Sandbox to live corporate number; run firebase deploy --only functions.